As is the case for Newton iteration, one may use the constraint residuals as an additional test. It is certainly unsafe to check only the latest difference of iterates (of a single component), and one should instead check that these are small for an entire cycle. The iteration described above is an example of a Gauss-Seidel-Newton iteration and it and many other iterative nonlinear solvers are discussed in [292].

4.3.4 Symplectic Constrained Dynamics Integrators

Is (4.14)–(4.16) symplectic (in the sense outlined above for holonomic systems)?

It is difficult to make sense of this question, since (4.14)–(4.16) does not even constitute a map of the co-tangent bundle. On the other hand it is not too difficult to correct this defect by incorporating an additional projection onto the cotangent space:

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\bar{\boldsymbol{p}}_{n+1}, \tag{4.20}$$

$$\bar{\boldsymbol{p}}_{n+1} = \boldsymbol{p}_n - h\boldsymbol{F}_n - h\boldsymbol{g}'(\boldsymbol{q}_n)^T\boldsymbol{\lambda}_n, \tag{4.21}$$

$$\boldsymbol{0} = \boldsymbol{g}(\boldsymbol{q}_{n+1}), \tag{4.22}$$

$$\boldsymbol{p}_{n+1} = \bar{\boldsymbol{p}}_{n+1} - \boldsymbol{g}'(\boldsymbol{q}_{n+1})^T\boldsymbol{\mu}_{n+1}, \tag{4.23}$$

$$\boldsymbol{0} = \boldsymbol{g}'(\boldsymbol{q}_{n+1})\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1}. \tag{4.24}$$

The demonstration that the method is symplectic can be seen as a consequence of the following lemma.

Lemma 4.1 *Let $\gamma : \mathbb{R}^{N_c} \to \mathbb{R}$ be C^2. Let $\boldsymbol{q}, \boldsymbol{p} \in \mathbb{R}^{N_c}$ be constrained to the manifold defined by*

$$\gamma(\boldsymbol{q}) = 0, \qquad \boldsymbol{p} \cdot \nabla\gamma(\boldsymbol{q}) = 0,$$

and suppose $\boldsymbol{P}, \mu$ satisfy

$$\boldsymbol{P} = \boldsymbol{p} - \mu\nabla\gamma(\boldsymbol{q}).$$

Then

$$\sum_{j=1}^{N_c} \mathrm{d}q_j \wedge \mathrm{d}P_j = \sum_{j=1}^{N_c} \mathrm{d}q_j \wedge \mathrm{d}p_j.$$

Proof Clearly the result follows provided

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_j \wedge \mathrm{d}\left[\mu \nabla \gamma(\boldsymbol{q})\right]_j = 0,$$

where $[\ldots]_j$ indicates the jth component of the term in brackets. We have

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_j \wedge \mathrm{d}\left[\mu \nabla \gamma(\boldsymbol{q})\right]_j = \sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_j \wedge \left(\frac{\partial \gamma}{\partial q_j} \mathrm{d}\mu + \mu \sum_{i=1}^{\mathrm{N}_c} \gamma''_{ij} \mathrm{d}q_j \right).$$

Then we note

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_j \wedge \frac{\partial \gamma}{\partial q_j} \mathrm{d}\mu = \sum_{j=1}^{\mathrm{N}_c} \frac{\partial \gamma}{\partial q_j} \mathrm{d}q_j \wedge \mathrm{d}\mu = 0,$$

since $\gamma(q) \equiv 0$ implies $\mathrm{d}\gamma(q) = \sum_{j=1}^{\mathrm{N}_c} \frac{\partial \gamma}{\partial q_j} \mathrm{d}q_j \equiv 0$. Moreover

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_j \wedge \sum_{i=1}^{\mathrm{N}_c} \gamma''_{ij} \mathrm{d}q_j = 0,$$

since the Hessian matrix γ'' is symmetric. □

Observe that

$$\boldsymbol{g}'(\boldsymbol{q}_{n+1})^T \boldsymbol{\mu}_{n+1} = \sum_{k=1}^{l} \mu_{n+1,k} \nabla g_k(\boldsymbol{q}_{n+1}).$$

Thus we may apply Lemma 4.1 with γ replaced by each g_k to demonstrate

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n+1,j} \wedge \mathrm{d}p_{n+1,j} = \sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n+1,j} \wedge \mathrm{d}\bar{p}_{n+1,j}.$$

Using similar argumentation based on the properties of the wedge product, we next show

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n+1,j} \wedge \mathrm{d}\bar{p}_{n+1,j} = \sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n,j} \wedge \mathrm{d}\bar{p}_{n+1,j},$$

and, finally, invoking Lemma 4.1 again,

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n,j} \wedge \mathrm{d}\bar{p}_{n+1,j} = \sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n,j} \wedge \mathrm{d}p_{n,j}.$$